Knowledge Organiser — 1.5 Legal, Moral, Cultural & Ethical Issues

OCR A Level Computer Science H446 — Component 01, Section 1.5 (1.5.1 legislation; 1.5.2 moral/ethical/cultural/social impact)

The four laws (1.5.1)

Act	What it covers	Key provision	Example breach
Data Protection Act 1998/2018 (+ GDPR)	How orgs collect, store & process **personal data** (data identifying a living individual)	7 principles: lawful/fair/transparent, purpose limitation, data minimisation, accuracy, storage limitation, integrity & confidentiality (security), accountability. Roles: **controller / processor / subject**. Enforced by **ICO** (fines up to €20m or 4% turnover)	Selling customer data without a lawful basis; storing records unencrypted; keeping data longer than needed
Computer Misuse Act 1990	**Unauthorised access** to computer systems (hacking, malware)	**3 offences:** 1 unauthorised access; 2 access **with intent** to commit further offence; 3 unauthorised **modification** (malware/ransomware/deleting). About *authorisation*, not damage	Employee using manager's login (1); hacking a bank to steal money (2); releasing ransomware (3)
Copyright, Designs & Patents Act 1988	**Intellectual property** — original works, software, music, designs	Protection is **automatic** (no registration); rights-holder controls copying/distributing/adapting; need a **licence** to use. Protects *expression*, not the idea	Pirating software/music; copying source code without a licence; using web images commercially without permission
Regulation of Investigatory Powers Act 2000 (RIPA)	How **public bodies** (police, security services) monitor/intercept communications	Allows lawful interception for national security/crime; ISPs must give access; can demand encryption keys; sets **who may authorise** to prevent abuse	A public authority intercepting communications **without proper authorisation** or for an improper purpose

Picking the right law: outsider hacks in = **CMA**; org mishandles data it holds = **DPA**; copying works = **CDPA**; state interception = **RIPA**. *Two can apply* (e.g. hacker steals data = CMA, from a firm that stored it unencrypted = DPA). Patents (vs copyright) = inventions, **must be applied for & granted**.

Moral / ethical / cultural / social issues (1.5.2)

Each is a trade-off — the same fact benefits one stakeholder and harms another.

Issue	One-liner	Benefit (+)	Drawback (−)
Automation & workforce	Machines do tasks once done by humans	Productivity, lower cost, new jobs created	Job displacement of low-skilled; deskilling
Automated decision making	Computers decide (loans, CVs, self-driving)	Fast, consistent, handles huge data, no fatigue	No empathy/judgement; hard to appeal; unclear liability (trolley problem)
AI ethics / bias	AI learns from data and acts autonomously	Powerful pattern-finding, efficiency, diagnosis	Reproduces/amplifies **bias**; "black box"; accountability unclear
Environmental	Computing's footprint and its green uses	Cuts travel (remote work), smart grids, paperless	E-waste/toxins, huge data-centre & crypto energy, rare-earth mining
Censorship & internet	Controlling what can be accessed/published	Protects children, blocks illegal/extremist content	Restricts free speech; abused by states to silence dissent
Monopolies & open-source	One firm dominating a market	Standards, polish, support (proprietary)	Stifles competition, high prices, **vendor lock-in** (open-source counters this)
Offshoring	Moving work abroad to cut costs	Cheaper labour/products, "follow the sun", global talent	Local job losses; quality/time-zone/data-security concerns
Layoffs	Mass redundancies from automation/offshoring	Lower costs, leaner, more competitive firm	Unemployment, lost income, regional/social impact
Digital divide	Gap between those with/without tech access & skills	Online services, education, jobs for the connected	Excludes poor/rural/elderly/disabled — widens inequality
Privacy & surveillance	Mass data collection by firms & governments	Personalised services, crime prevention, public health	Loss of privacy, profiling, breaches, chilling effect, misuse (links DPA/RIPA)

Extended-response (essay) technique — for 9–12 mark "discuss/evaluate"

Skeleton: FOR → AGAINST → FOR → AGAINST → justified conclusion. For each point: *point → consequence → stakeholder → counter-point.*

- **Name multiple stakeholders** (aim 3–4+): individuals/consumers/employees · businesses/employers · government/society · the environment · developing countries/future generations.
- **Levels of response** (whole-answer banding, not point-by-point):
- **Level 1 (low):** fragmented one-sided list, no development, no conclusion.
- **Level 2 (mid):** some discussion but mostly descriptive/one-sided; weak/absent conclusion.
- **Level 3 (top):** balanced, multiple stakeholders, **both** benefits & drawbacks, accurate terms, sustained reasoning, **justified conclusion** that takes a position.

- **Conclusion must take a side and justify it** — often with a condition (e.g. "automate, *but only if* staff are retrained and some staffed tills remain"; "surveillance only if targeted, proportionate & independently authorised").
- **AOs:** AO1 recall the law · AO2 apply to scenario · AO3 evaluate. Essays are mostly AO2/AO3.

Top exam reminders

- **Never one-sided.** "Discuss/evaluate" needs both sides + a justified conclusion to reach Level 3.
- **Always conclude** — take a clear, reasoned position; don't just stop.
- **Develop, don't list.** "It causes job losses" → why, who's affected, what follows, counter-argument.
- **Name the specific Act/offence** — not just "it's illegal" (e.g. CMA *Offence 3*). Note when **two laws** apply.
- **GDPR ≠ DPA 2018:** GDPR is the EU regulation; DPA 2018 is the UK law implementing it.